DESIGN AND ANALYSIS OF ALGORITHM

LAB EXPERIMENT -8

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ROLL NO:

CH.SC.U4CSE24242

1. HUFFMAN CODING CODE:

```c
#include <stdio.h>
#include <stdlib.h>
#define MAX 100
struct node {
    char data;
    int freq;
    struct node *left, *right;
};
struct node* createNode(char data, int freq) {
    struct node* temp = (struct node*)malloc(sizeof(struct node));
    temp->data = data;
    temp->freq = freq;
    temp->left = temp->right = NULL;
    return temp;
}
void printCodes(struct node* root, int arr[], int top) {

    if (root->left) {
        arr[top] = 0;
        printCodes(root->left, arr, top + 1);
    }

    if (root->right) {
        arr[top] = 1;
        printCodes(root->right, arr, top + 1);
    }

    if (!root->left && !root->right) {
        printf("%c: ", root->data);
        for (int i = 0; i < top; i++)
            printf("%d", arr[i]);
        printf("\n");
    }
}
int main() {
    printf("\n roll no: CH.SC.U4CSE24215");
    printf("\n Name : Hemal");
    char data[] = {'a','b','c','d','e','f'};
    int freq[] = {5,9,12,13,16,45};
    int n = 6;
    struct node* nodes[MAX];
    for(int i=0;i<n;i++)
        nodes[i] = createNode(data[i], freq[i]);
    while(n > 1) {
        int min1 = 0, min2 = 1;
        if(nodes[min1]->freq > nodes[min2]->freq){
```

```
            int temp = min1;
            min1 = min2;
            min2 = temp;
        }

        for(int i=2;i<n;i++){
            if(nodes[i]->freq < nodes[min1]->freq){
                min2 = min1;
                min1 = i;
            }
            else if(nodes[i]->freq < nodes[min2]->freq)
                min2 = i;
        }

        struct node* left = nodes[min1];
        struct node* right = nodes[min2];
        struct node* newNode = createNode('$', left->freq + right->freq);
        newNode->left = left;
        newNode->right = right;
        nodes[min1] = newNode;
        nodes[min2] = nodes[n-1];
        n--;
    }
    int arr[MAX], top = 0;
    printf("Roll no : ch.sc.u4cse24215\n");
    printf("Name : Hemal\n");
    printf("Huffman Codes:\n");
    printCodes(nodes[0], arr, top);
    return 0;
}
```

OUTPUT:

```
hemal@Admin:~$ nano huffman.c
hemal@Admin:~$ gcc huffman.c -o huffman
hemal@Admin:~$ ./huffman

 roll no: CH.SC.U4CSE24215
 Name : HemalRoll no : ch.sc.u4cse24215
Name : Hemal
Huffman Codes:
f: 0
c: 100
d: 101
a: 1100
b: 1101
e: 111
hemal@Admin:~$
```

INTERPRETATION:

Huffman coding is a greedy algorithm used for data compression by assigning shorter binary codes to characters with higher frequencies. It builds a binary tree by repeatedly combining the two nodes with the smallest frequencies until a single tree is formed.